

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
SHINYANGA REGION
FORM FOUR MOCK EXAMINATION - MAY 2024
PHYSICS 2A: SUBJECT CODE: 031/2A

TIME: 2:30HRS DATE: MAY, 2024

- 1. This examination has only one (1) section.**
- 2. Answer ALL questions.**
- 3. Write your examination number on every page of your answer sheet/
booklet provided.**
- 4. Cellular phones are not allowed in the examination rooms.**
- 5. Non-programmable calculator and mathematical table are allowed in
the examination room.**
- 6. Where necessary the following constant may be used.**
 - i). Density of water = $1\text{g/cm}^3 = 1000\text{kg/m}^3$.**
 - ii). Acceleration due to gravity $g = 10\text{m/s}^2$.**

GEPAM science hub

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1. You are provided with the following apparatus:-

Measuring cylinder, A eureka can, nylon thread, standard masses and water.

Proceed as follows:-

- Pour water into eureka can until it just beginning to overflow.
- Hold a suitable measuring cylinder under the spout and immerse a standard mass of 50g into eureka can as shown in figure 1 below.
- Wait for it to drop until it starts to cease and take long interval to drop.

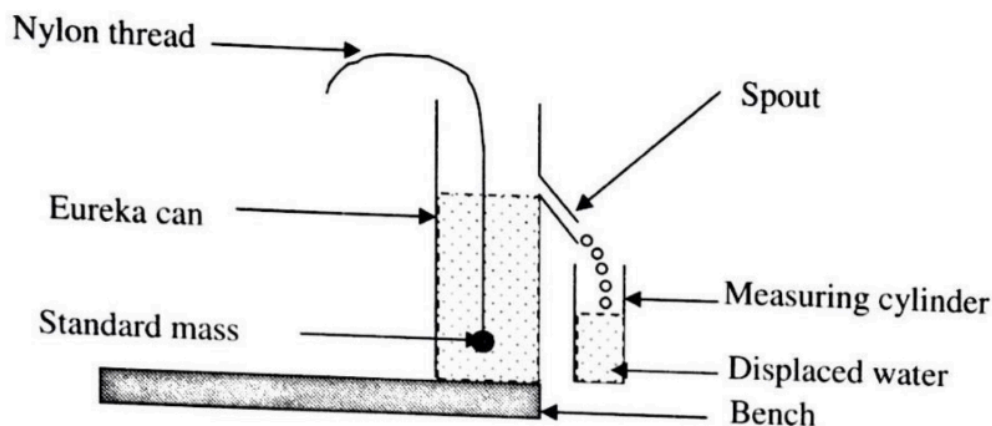


Figure 1

- Record the reading of the water collected in the measuring cylinder.
- Repeat the procedures (b) to (d) for standard masses 100g, 150g, 200g and 250g.
- Tabulate your results showing the quantities as follows:-

Mass, $M(\text{g})$	50	100	150	200	250
Volume, $V(\text{cm}^3)$					

- Plot a graph of mass (M) against Volume (V).
 - From the graph:-
 - Calculate the slope and state its significance.
 - Establish the formula governing the experiment.
 - Identify with reasons the best to the least satisfactory method of finding the constant value of mass divided by volume.
 - State two possible errors in this experiment.
 - How can you minimize the errors you have just mentioned?
2. You are provided with a source of power 3V, Rheostat, Ammeter, Resistor, Voltmeter, key and connecting wires, set the apparatus as shown in figure 2.

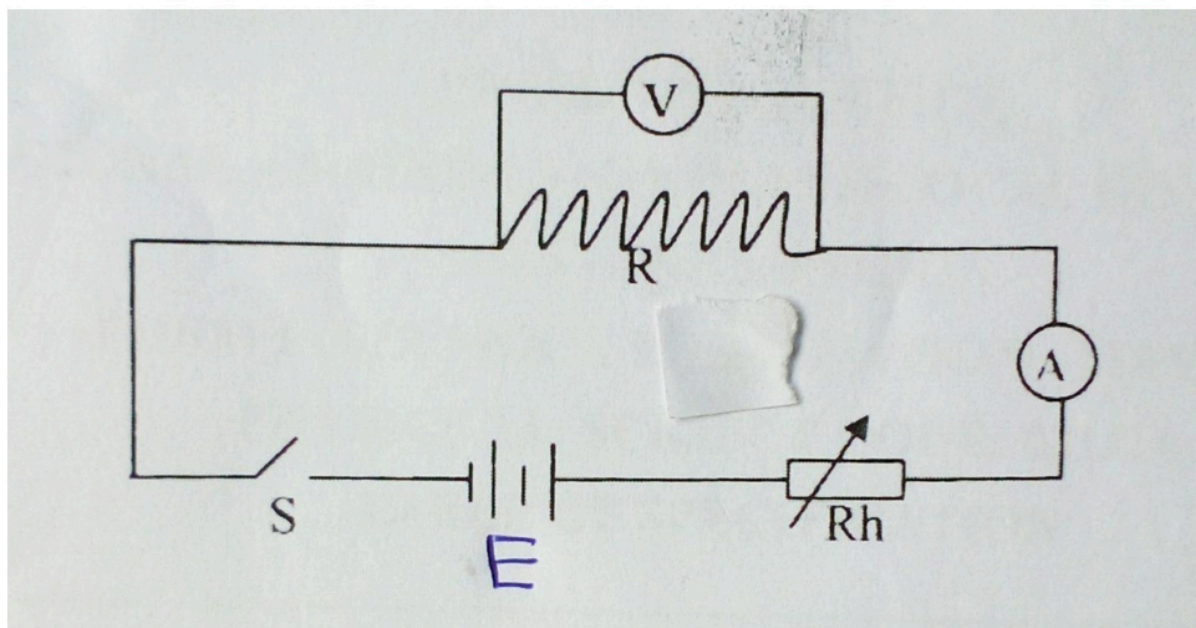


Figure 2

- a) Name the special symbols used in the circuit in figure 2.
- b)
 - i) Close the switch and adjust the R_h by sliding slowly from one end so that the ammeter reads the maximum value.
 - ii) Read and record the values of V and I the position of slider on R_h for five (5) different positions and tabulate your results.
- i) Plot a graph of V against I .
 - ii) What is the shape of your graph?
 - iii) Determine the slope of your graph.
 - iv) What is the physical meaning of the slope obtained in 2(c) (iii)?
 - v) From the graph, determine the value of p.d when the current is 0.15A.
 - vi) What is the relationship between V and I across R ?
 - vii) Why is this experiment not an accurate method of calculating resistance?
 - viii) State the law governing this experiment?
- c) What is the aim of this experiment?
- d) State one source of error and how to minimize it.

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MARKING SCHEME

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1. (f) Table of results

Mass M (g)	50	100	150	200	250
Volume V(cm³)	06	14	20	26	32
Error ± 0.5	01	01	01	01	01

(g) The graph of M (g) against V (cm³) is drawn on the graph paper.

(h) Slope (G) = $\frac{\Delta M(g)}{\Delta V(cm^3)}$

$$G = \frac{(160 - 70g)}{(21 - 6)cm^3}$$

$$G = \frac{90g}{15cm^3}$$

$$G = 6g/cm^3$$

The slope of a graph is 6g/cm³.

i. The slope of a graph represents the density of a standard mass (iron used)

ii. The formula governing the experiment is density (ρ) = $\frac{Mass(g)}{Volume(cm^3)}$

(i) The best method is to find the slope of a graph which is an average of the ratio of mass divided by volume.

(j) Sources of errors

- ✓ Standard masses having different densities which lead to having different sizes.
- ✓ Size of a string used. Thick string leads to inaccurate reading of volume displaced.
- ✓ Graph caused errors.
 - i. Use of blunt pencil in drawing graph.
 - ii. Too much estimation when drawing graph.

(k) Ways to minimize errors

- ✓ Use identical standard masses.
- ✓ Use a thin string.
- ✓ Use a sharp pencil
- ✓ Transfer the data correctly, avoid too much approximation.

2. (a) A – Ammeter
 V – Voltmeter
 E – Battery
 Rh – Rheostat
 R = Resistor
 K = Key

(b) (iii) Table of results.

Voltage V(Volt)	02.22	1.80	1.60	1.40	0.40
Current I(Ampere)	0.55	0.50	0.45	0.40	0.10

(c) (i) The graph of V against I is plotted on the graph paper.

(ii) The shape of a graph is a straight line graph.

(iii) Slope (G) = $\frac{\Delta V(V)}{\Delta I(A)}$

$$G = \frac{(2.15 - 0.9)V}{(0.6 - 0.25)A}$$

$$G = \frac{1.25}{0.35}$$

$$G = 3.57\Omega$$

(d) (iv) The meaning of a slope is the value of resistance of a resistor R which is 4Ω

(v) From the graph the value voltage is 0.52V.

(vi) Current (I) is directly proportional to voltage (V).

(vii) Because is not applicable in electrolytes, semiconductors.

(viii) The law is called Ohm's law.

Ohm's law states "The current flowing through a conductor is directly proportional to the potential difference across it and provided that temperature and other physical factors remain constant"

(e) The aim of the experiment was to verify Ohm's law or to find the value of unknown resistance R.

(f) **Sources of error.**

- ✓ Instrumental errors. Such as voltmeter, Ammeter, Rheostat not working properly.
- ✓ Parallax errors, due to poor observation on recording voltage and current.

Minimizing error

- ✓ Use the accurate/well calibrated instrument.
- ✓ Position properly the eyes during recording data.